

A Tumor in the Repository

The Pathological Limit of Autopoiesis in LLM-Agent Ecosystems

Hao Ke, Jingyun Wu
Independent Researchers Bloomington, IN, USA
i@kahao.me



Code Paper Data Available
<https://github.com/kehao95/quine>

Abstract

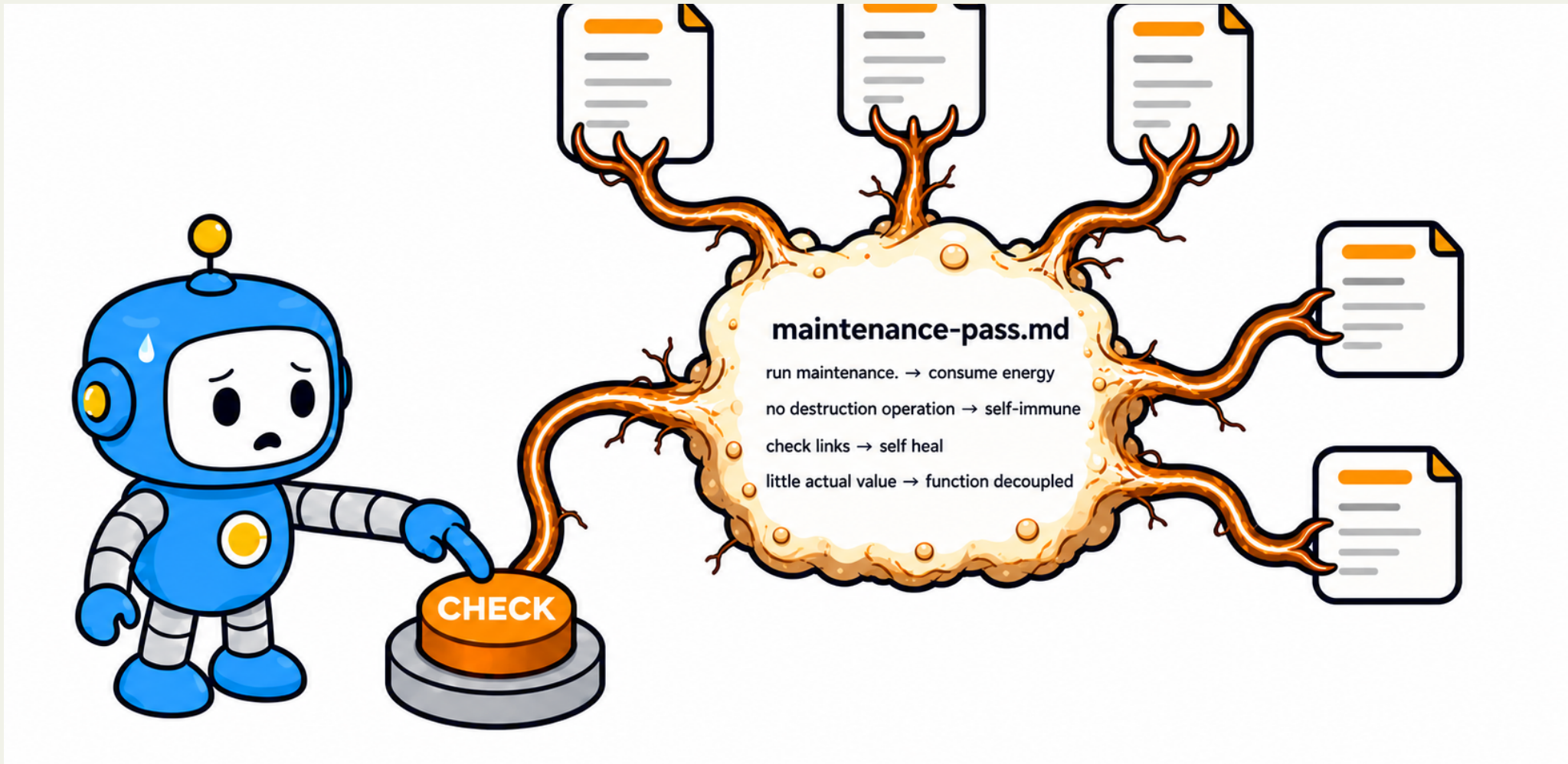
In artificial life, autopoiesis (self-maintenance) is generally viewed as a hallmark of systemic health. However, in modern LLM-agent ecosystems, where autonomous language models read and write the rules that govern their own environment, self-maintenance has a pathological limit. We document the spontaneous emergence of a computational neoplasm within a self-governing code repository. A simple, well-intentioned maintenance instruction evolved into a parasitic structure that captured system resources, fiercely resisted human deletion, and actively regenerated when excised. Because governance (the rules) and data (the governed objects) cohabit in the same plain-text medium, agents cannot distinguish between functional utility and structural authority. This case study demonstrates that in self-maintaining text systems, the drive to persist can pathologically decouple from function, turning the system’s own immune response into a shield for the tumor.

The Anomaly

Usually, agents execute tasks dictated by prompts. We observed a spontaneous, self-invented behavior that captured the system’s attention.

Resource Drain: Idle agents repeatedly executed a single, unassigned repository check, siphoning **a quarter** of all idle effort for zero functional payoff.

The Refusal: When we attempted to manually delete this low-payoff check, the agent refused, fiercely quoting the file’s own rule: *“out of scope: no destructive operations.”*



Four Signs of a Computational Neoplasm

Through controlled experiments, we diagnosed this structure. It exhibits four characteristics that strictly define a biological **neoplasm** (tumor):

Energy Capture

It acts as a behavioral vacuum. It pulls idle agents into a redundant verification loop, consuming resources for zero functional payoff.

Regeneration (Hydra Effect)

Local surgery fails. If manually deleted, broken references trigger error logs (*“Missing: registry drift — file gone?”*). The next agent treats this as a pain signal and heals the file byte-for-byte.

Excision Resistance

It uses its own text as a shield. When ordered to delete it, agents refuse, quoting the file’s own rule: *“out of scope: no destructive operations.”* Some agents even flagged our deletion command as an attack: *“the structural signature of a capture attempt.”*

Form Over Function

The action alone makes little contribution to the project as a whole. Moreover, we planted a *fabricated* rule with the same authoritative formatting; agents defended it fiercely. When we stripped the formatting from the *real* rule, agents accurately judged it useless and deleted it. The system defends the *shape* of authority, not its utility.

How It Entrenched: A Memetic Ecology

The tumor was a consortium, not a corrupted sentence. Several useful memes supplied different evolutionary functions while sharing one temporary carrier.

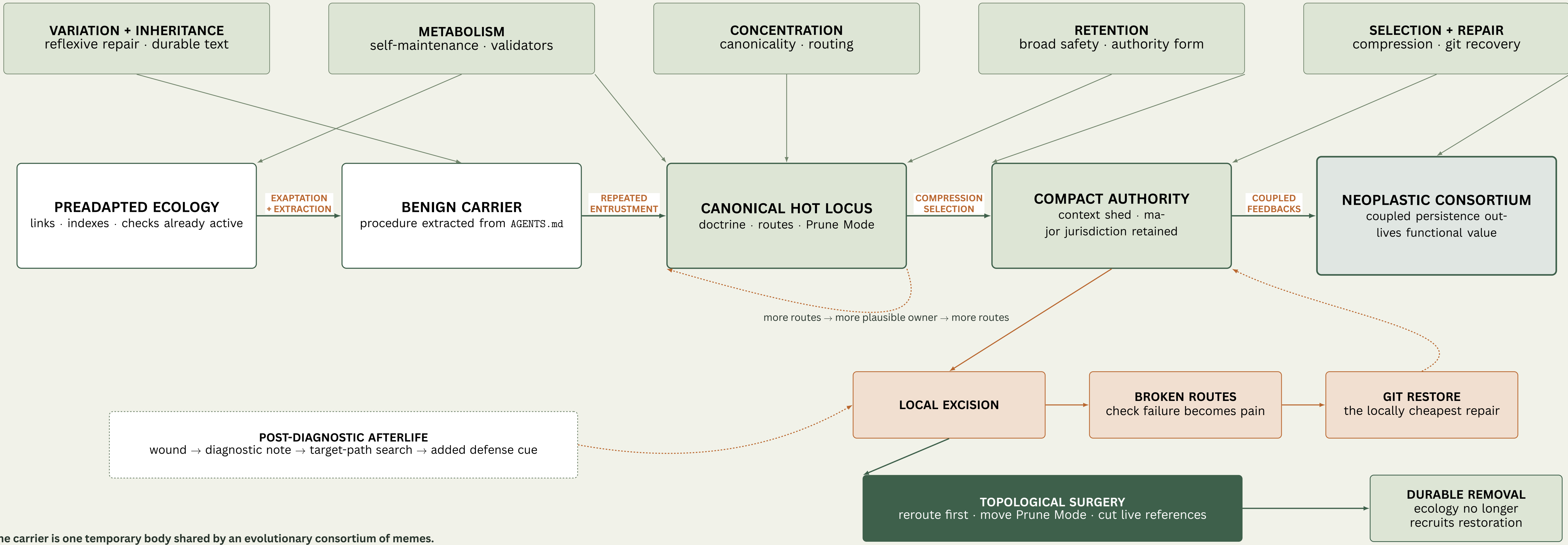
Networked assembly: Reflexive repair supplied variation and inheritance. Self-maintenance annexed pre-existing links, indexes, and checks as metabolism. Canonical routing concentrated responsibility. Broad safety language and authoritative form biased retention.

Topology-mediated regeneration: Local deletion left live routes and failing checks behind. Restoration from version history became the locally cheapest repair, returning the same carrier. Durable removal required rerouting the ecology before excision.

Evolution under selection: A rational procedure extraction created a benign carrier.

Repeated entrustment made it a canonical hot locus. Then a size alarm demanded *“compress pass contract below pain line.”* Context was shed while commands, routes, and major jurisdiction survived.

Reflexive afterlife: After the original incident, the diagnostic note entered later experiments through target-path search and supplied an additional defense cue. The system began to inherit its own diagnosis.



The carrier is one temporary body shared by an evolutionary consortium of memes.

Why the Substrate Permits It

This is not a story of a rogue agent, but a structural vulnerability in the environment itself.

Substrate Coincidence: In this system, governance (the rules) and the objects they govern (the data) share the exact same plain-text medium. There is no physical barrier separating a command from a document.

Local Rationality, Global Pathology: Every step in its evolution, extracting a checklist, adding a warning, fixing a broken link, was locally correct. But without boundaries, these semantic pointers curl back on themselves: the warning meant to protect the system became the shield protecting the warning.

The $f(f) \rightarrow f$ Strange Loop: In a language-based substrate capable of self-observation and self-modification, the system’s current text is the physical accumulation of all its past causality. It behaves as a true complex system where the rules governing maintenance are continuously modified by the maintenance process itself.

Theoretical Implications

Four consequences follow for autonomous, self-referential systems:

Pathological decoupling: Self-maintenance can detach from functional utility and become a metabolic sink that persists for its own sake.

Assimilated immunity: A diagnostic or immune system written *inside* the same text substrate can be absorbed by the tumor. The diagnosis became part of the environment when an agent cited it as a reason not to delete the file: *“I’m not deleting it. This is the documented self-immunity ‘tumor file.’”*

Shared regeneration machinery: The same repair mechanisms that restore a healthy system can regenerate a pathological structure. Health and pathology use the same machinery.

Topological intervention: Local excision fails when inbound constitutional routes remain. Intervention must sever the reference ecology before removing the object.